

CSE630, SUST, 7 Aug 2025

Name:

Duration: 1 hour

Reg:

Marks: 20

Write your answers directly on the question paper.

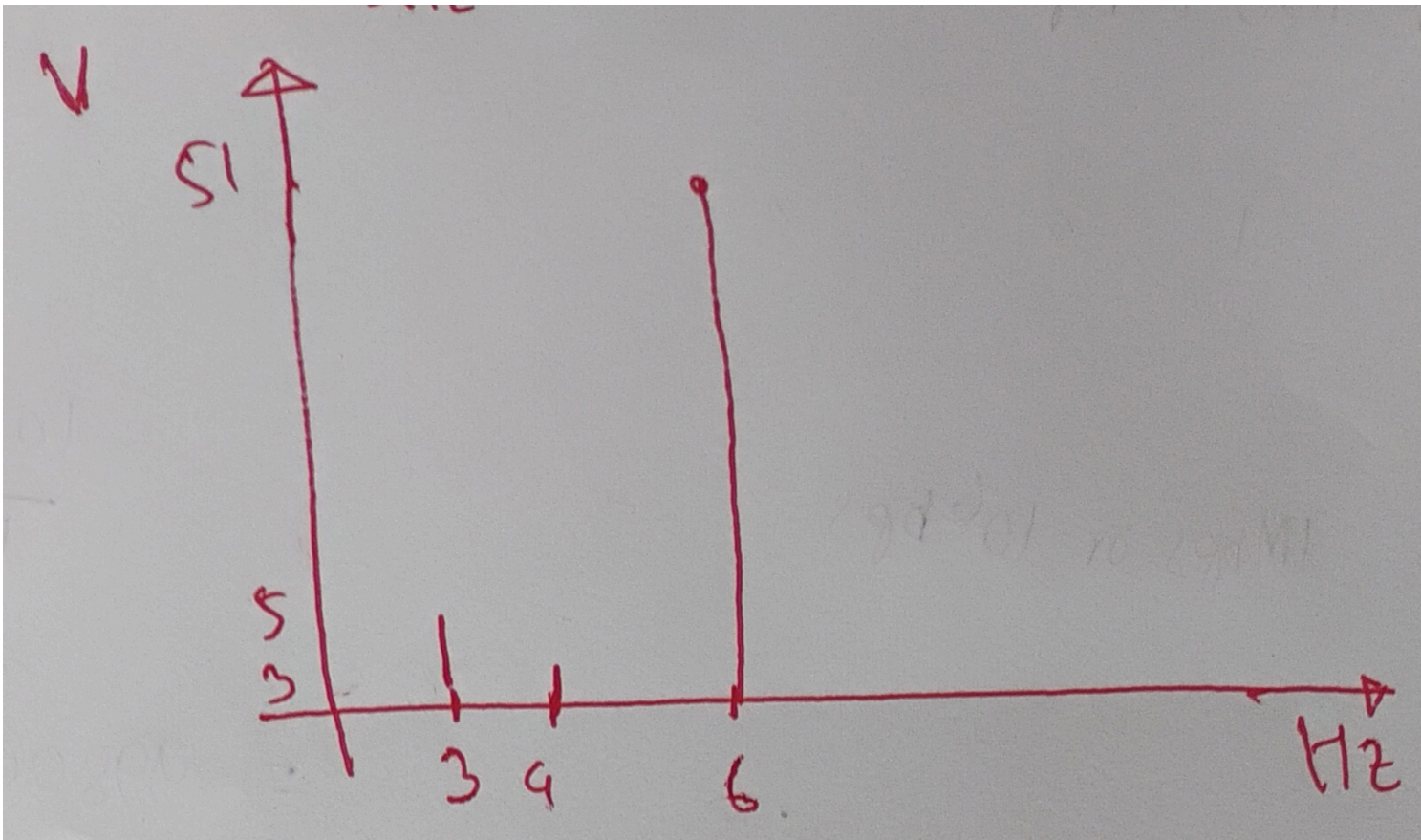
1. Packet can take multiple paths from sender to receiver. Which layer in the OSI model is responsible for determining the best path? (.5) Network
2. If a PDU is lost in the network, which layer in the OSI model is responsible for recovering it? (.5) Transport
3. When a sender transmits data at an excessive rate, the receiver can become overwhelmed and unable to handle other communications. Which layer in the OSI model prevents this situation? (.5) Transport
4. Data needs to be encrypted so that any man in the middle cannot understand the conversation. Which layer in OSI model is responsible for the encryption? (.5) Presentation
5. Provide the correspondence between the layers of the OSI model and the TCP/IP model. (.5)

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet / Network
Data Link	Network Access / Network Interface
Physical	

6. There is a layer in the OSI model that divides the data to be sent into chunks.

a. Which layer is it? (.5) Transport

b. When the receiver receives the data, it assembles the chunks into the complete data. What does the layer do to ensure the receiver knows the correct order of the chunks? (.5) Sequence Number
7. A composite sine wave has three sine waves: one with frequency 3 Hz with amplitude 5 V, one with frequency 4 Hz with amplitude 3 V and last one with frequency 6 Hz with amplitude 51V. Draw the signal in frequency domain. (1)



8. Sender can send at 1 Mbps and the delay of the link is 5s.

a. What is the maximum number of bits that can fill the link. (1.5)

delay X rate = 5 Mb

b. The link is full-duplex. The sender sends a burst of data of bits and waits for receiver acknowledgment for part of the burst before sending another burst. Determine the maximum burst size. (1.5)

2 X delay X rate = 10 Mb

8. If digital data is already in digital format, why does it still need to be converted into a digital signal for transmission? Write points only. (1)

Baseline Wandering

DC Components

Self-synchronization

Built-in Error Detection

Immunity to Noise and Interference

Complexity

9. The attenuation of a signal is -10 dB. What is the final signal power if it was originally 5 W? (1)

$$\begin{aligned}
 \text{dB} &= 10 \log \frac{P_2}{P_1} \\
 \Rightarrow -10 &= 10 \log \frac{P_2}{5} \\
 \Rightarrow \log \frac{P_2}{5} &= -1 \\
 \Rightarrow \frac{P_2}{5} &= 10^{-1} \\
 \Rightarrow P_2 &= \frac{5}{10} = .5 \text{ W}
 \end{aligned}$$

10. A signal has passed through three cascaded amplifiers, each with a 4 dB gain. What is the total gain? How much is the signal amplified? (1)

4 dB + 4 dB + 4 dB = 12 dB

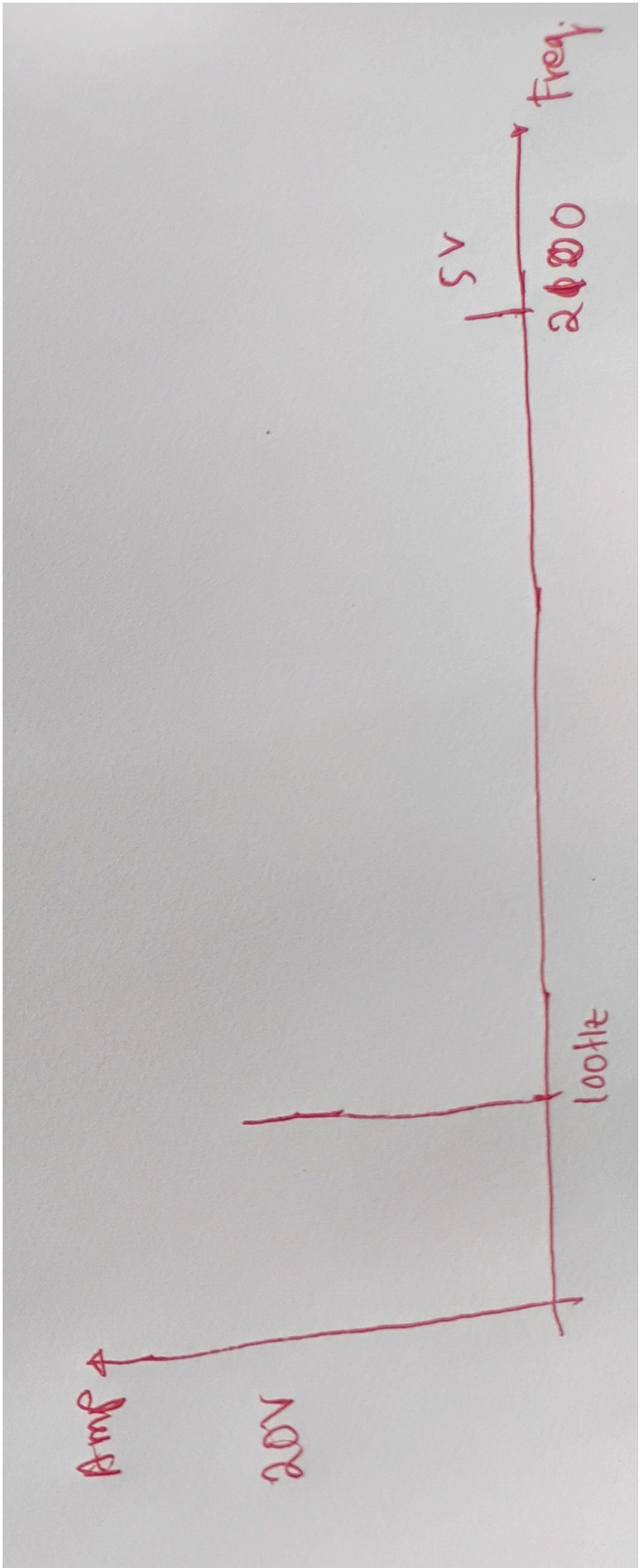
11. We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level? (1.5)

$$C = B \log_2(1 + \text{SNR}) = 6 \text{ Mbps}$$

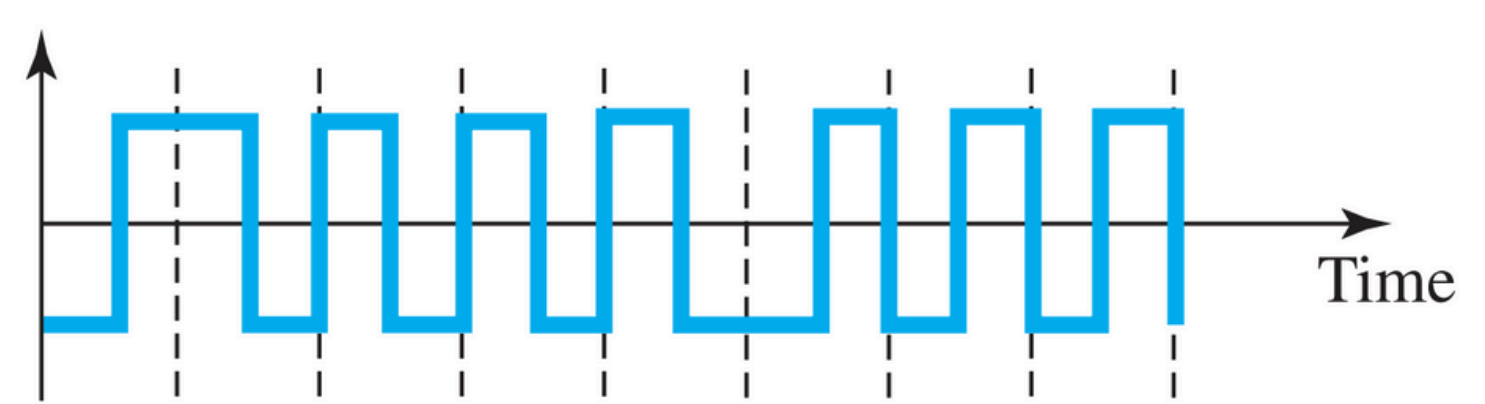
$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L \Rightarrow L = 4$$

12. A periodic composite signal with a bandwidth of 2000 Hz is composed of two sine waves. The first one has a frequency of 100 Hz with a maximum amplitude of 20 V; the second one has a maximum amplitude

of 5 V. Draw the bandwidth. (1.5)

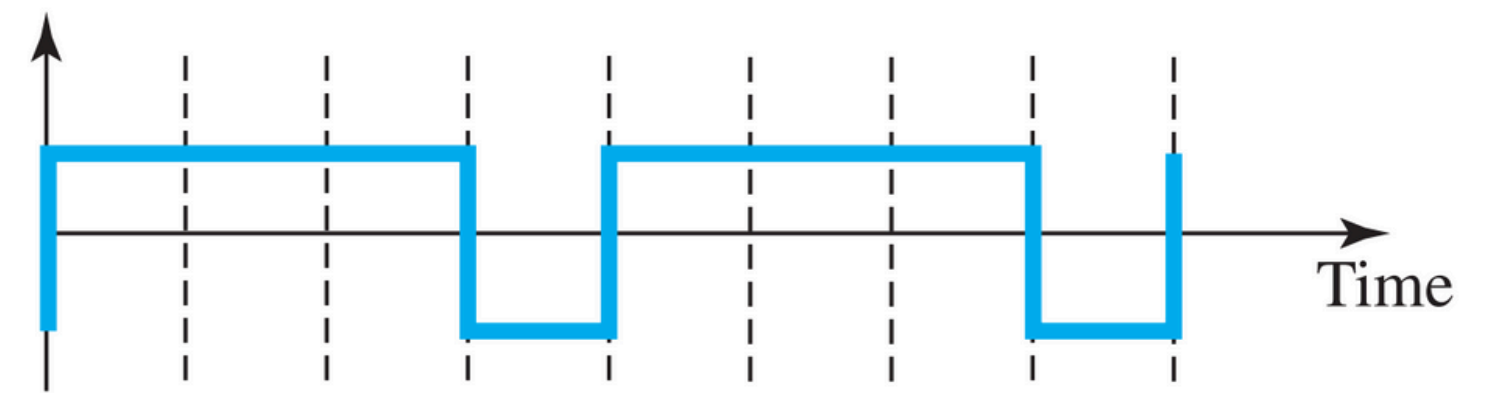


13. Find the 8-bit data stream for differential Manchester. Assume the signal level is low immediately before the start of the signal. (1.5)



11000100

14. Find the 8-bit data stream for NRZ-I. Assume the signal level is low immediately before the start of the signal. (1.5)



10011001

15. Write down all the layers of OSI model, their tasks, and addressing if available. (2)

OSI	Tasks	Addressing
Application	Name System DNS Host Config BOOTP, DHCP Email SMTP, POP, IMAP File Transfer FTP, TFTP Web HTTP, HTTPS	Specific Address
Presentation	Translation Compression Encryption	
Session	Session Management Authentication Authorization Synchronization	
Transport	Connection Control Flow and Error Control Segmentation and Reassembly Multiplexing	Port Address (16 bit)
Network	Providing Communication Channel Routing	Logical address (IP, 32 bit)
Data Link	Media Access Control	Physical Address (MAC, 48 bit)
Physical	Providing Transmission in Communication Line	

16. In a digital transmission, the sender clock is 0.2 percent faster than the receiver clock. How many extra bits per second does the sender send if the data rate is 1 Mbps? (1.5)

sender's clock is .2 percent faster than the receiver's

Receiver: 100 bps

Sender: 100.2 bps

Sender send 100.2 bps and receiver ~~receives~~ received 100 bps

" " " " " "

" " 1 Mbps or 10^6 bps

$$\frac{100}{100.2} \text{ bps}$$

$$\frac{100 \times 10^6}{100.2} \text{ bps}$$

$$= 998003.99$$

$$\approx 998004$$

$$\text{So Ans } 10^6 - 998004 = 1996$$